

Sidharth Sreeram

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University of Waterloo Mechatronics Engineering graduate specializing in AI/ML, with experience fine-tuning multimodal vision-language models and building deep learning pipelines in PyTorch. Skilled in data pipeline design, model evaluation, and applying ML research to real-world perception problems.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL, Java, TypeScript, XML

Frameworks & Libraries: PyTorch, Hugging Face TRL, Scikit-Learn, OpenCV, NumPy, Pandas, ROS2

Concepts: LLM, VLM, VLA, Computer Vision, Neural Networks, Reinforcement Learning, Transformers

Development: Linux, Databricks, Git, Oracle Cloud, Docker, GitHub, Jira, Confluence, Agile

EDUCATION

University of Waterloo - *B.ASc in Mechatronics Engineering, Specializing in Artificial Intelligence*

EXPERIENCE

Research Assistant - UW WISE Lab | *Python, PyTorch* Jan – Apr 2026

- Developed Python pipelines to structure data into standardized VLM prompt formats for instruction tuning.
- Designed prompt templates from the OpenDriveVLA publication for standardized scene-data instructions.
- Surveyed Visual QA dataset literature to inform autonomous-driving instruction-tuning data design.

Software Engineering Analyst - Tamaki Controls | *Python, Java, SQL* May – Aug 2025

- Designed an API response schema that improved cross-service communication between Java and Python.
- Engineered backend pipelines generating CSV/JSON outputs for direct use in analytics and ML workflows.
- Used UTF-8 encoded CSV/JSON byte streams to improve memory efficiency and reduce encoding ambiguity.
- Reviewed and updated API documentation with clear input/output descriptions and example Python scripts.

Platform Software Engineer - Ford Canada | *Python, C++, GCov* Sept – Dec 2024

- Developed a Python CLI for the AUTOSAR State Management API in a containerized Linux environment.
- Automated state flow logic to evaluate runtime conditions and perform transitions automatically.
- Enabled testers and integrators to easily control system states by eliminating manual navigation of state architecture.
- Created C++ test suites, increased line coverage from 45% to 94% and condition coverage from 50% to 85%.

Controls System Engineer - Brock Solutions | *Python* Jan – Apr 2024

- Built human-machine interfaces for integrated real-time updates and visualized plant processes.
- Engineered process simulations to replicate full plant behavior, enabling rapid validation of HMI designs.
- Wrote scripts to generate synthetic time-series data and accelerated state transitions.

PROJECTS

Qwen 3 VQA Model Tuning | *Python, PyTorch, Qwen 3, VLM, LoRA, Hugging Face* [GitHub Repo ↗](#)

- Fine-tuned Qwen 3 to perform driving scene QA reasoning using camera and LiDAR data from the NuScenes dataset.
- Built a training pipeline using LoRA, quantization, and HuggingFace TRL to perform under limited GPU resources.

Snake Game Reinforcement Learning | *Python, PyTorch, Deep Q-Networks* [GitHub Repo ↗](#)

- Modeled a Deep Q-Network to learn optimal control policies for the snake game using reinforcement learning.
- Implemented epsilon-greedy exploration and discounted future rewards to stabilize and improve performance.

Path Following Robot | *Python, Raspberry Pi, OpenCV, YOLOv8, Git* [GitHub Repo ↗](#)

- Designed and implemented a real-time computer vision system for color-based object detection and path following.
- Applied a camera homography transformation, enabling accurate navigation and target localization.
- Proposed and tested a YOLOv8 object detection model to notify upon locating the target.